

Yellevate

Work Documentation

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Raw Data (Excel)

A snapshot of the raw data of customer invoices in Excel, subject to cleaning and transformation.

	A	B	C	D	E	F	G	H	I	J	K	L
1	country	customer_ID	invoice_number	invoice_date	due_date	invoice_amount_usd	disputed	dispute_lost	settled_date	days_to_settle	days_late	
2	China	0379-NEVHP	611365	02/01/2021	01/02/2021	5594	0	0	15/01/2021	13	0	
3	France	2621-XCLEH	6482427308	13/01/2020	12/02/2020	8099	1	0	14/03/2020	61	31	
4	China	2820-XGXS8	9231909	03/07/2021	02/08/2021	6588	0	0	08/07/2021	5	0	
5	France	9322-YCTQO	9888306	10/02/2021	12/03/2021	10592	0	0	17/03/2021	35	5	
6	Russia	6627-ELFBK	15752855	25/10/2020	24/11/2020	7227	1	0	28/11/2020	34	4	
7	Russia	5148-SYKLB	18104516	27/01/2020	26/02/2020	9400	1	0	22/02/2020	26	0	
8	Spain	8690-EEBEO	23864272	13/08/2021	12/09/2021	7469	0	0	09/09/2021	27	0	
9	United Stat	4460-ZXNDN	27545037	16/12/2020	15/01/2021	7506	0	0	12/01/2021	27	0	
10	United Stat	3831-FXWYK	28049695	14/05/2020	13/06/2020	8007	1	0	01/07/2020	48	18	
11	Spain	7654-DOLHO	32277701	01/07/2021	31/07/2021	4833	0	0	26/07/2021	25	0	
12	United Stat	3993-QUNVJ	35868002	31/03/2020	30/04/2020	7533	0	0	16/04/2020	16	0	
13	France	5284-DJOZO	36478577	07/08/2021	06/09/2021	7335	0	0	15/08/2021	8	0	
14	Russia	5924-UOPGH	36620839	08/05/2021	07/06/2021	9008	1	0	09/06/2021	32	2	
15	France	9117-LYRCE	41324194	21/10/2020	20/11/2020	5717	0	0	30/11/2020	40	10	
16	Russia	7695-NKUXM	42511106	07/11/2020	07/12/2020	5002	0	0	18/11/2020	11	0	
17	China	8820-BLYDZ	46372811	21/02/2021	23/03/2021	6196	0	0	28/02/2021	7	0	
18	France	8976-AMJEO	46937392	16/05/2021	15/06/2021	6988	0	0	04/06/2021	19	0	
19	Russia	5148-SYKLB	49331333	29/05/2021	28/06/2021	6880	1	0	10/07/2021	42	12	
20	Russia	3568-JIMFW	52765186	20/10/2021	19/11/2021	9623	1	0	09/11/2021	20	0	
21	Russia	5613-UHVMG	55416013	30/11/2020	30/12/2020	4201	1	0	16/01/2021	47	17	
22	France	6833-ETVHD	57081728	04/11/2020	04/12/2020	8665	0	0	29/11/2020	25	0	
23	United Stat	7758-WKLVN	57781566	24/06/2020	24/07/2020	7149	0	0	07/08/2020	44	14	
24	Russia	6004-KITZM	58393139	06/01/2021	05/02/2021	8816	0	0	29/01/2021	23	0	
25	China	9841-XLGBV	72380981	16/02/2021	18/03/2021	2644	0	0	06/03/2021	18	0	
26	Spain	8690-EEBEO	75181247	18/02/2020	19/03/2020	8333	0	0	30/03/2020	41	11	
27	China	8156-PCYBM	81932735	24/01/2020	23/02/2020	7270	0	0	03/03/2020	39	9	
28	Russia	7946-HJDUR	86171934	29/05/2021	28/06/2021	4169	0	0	22/06/2021	24	0	
29	United Stat	3831-FXWYK	93006859	24/12/2020	23/01/2021	2446	0	0	31/01/2021	38	8	
30	France	3448-OWJOT	1297791719	15/01/2020	14/02/2020	4864	1	0	16/02/2020	32	2	
31	Russia	8887-NCUZY	101415601	31/05/2020	30/06/2020	4007	0	0	04/07/2020	34	4	
32	Spain	9286-VLKMI	101747306	27/05/2021	26/06/2021	5819	0	0	29/05/2021	2	0	
33	France	6833-ETVHD	103483331	27/07/2020	26/08/2020	7213	0	0	18/08/2020	22	0	
34	China	6160-HCSFI	104628267	13/01/2020	12/02/2020	7272	0	0	04/02/2020	22	0	
35	China	1080-NDGAE	106360977	19/02/2020	20/03/2020	9348	1	0	26/03/2020	36	6	
36	Spain	6296-UKEUZ	106486147	21/09/2021	21/10/2021	4655	0	0	25/09/2021	4	0	
37	United Stat	6831-FIODB	110122785	23/10/2020	22/11/2020	3258	0	0	30/11/2020	38	8	
38	Spain	6296-UKEUZ	120364375	23/04/2021	23/05/2021	3463	0	0	24/04/2021	1	0	
39	Russia	5148-SYKLB	121797094	28/09/2020	28/10/2020	8915	0	0	30/10/2020	32	2	

Dataset Information

The data that was extracted and analyzed:

- Country where the invoice was issued.
- Customer ID for each customer.
- Invoice amount in USD showing how much the customer needs to pay.
- Disputed where 1 means the customer disputed the

invoice and 0 means they did not.

- **DisputedLost** where 1 means Yellevate lost the dispute, and the customer opted out of paying the invoice. And 0 means they legally required to pay the full invoice amount, it either means that Yellevate won the dispute or there was no dispute at all.
- **DaysToSettle** is the number of days from when the invoice was generated to when it was settled.

	country character varying	customer_id character varying	invoice_number numeric	invoice_date date	due_date date	invoice_amount numeric	disputed numeric	dispute_lost numeric	settled_date date	days_settled integer	days_late integer
1	China	0379-NEVHP	611365	2021-01-02	2021-02-01	5594	0	0	2021-01-15	13	0
2	France	2621-XCLEH	6482427308	2020-01-13	2020-02-12	8099	1	0	2020-03-14	61	31
3	China	2820-XGXS	9231909	2021-07-03	2021-08-02	6588	0	0	2021-07-08	5	0
4	France	9322-YCTQO	9888306	2021-02-10	2021-03-12	10592	0	0	2021-03-17	35	5
5	Russia	6627-ELFBK	15752855	2020-10-25	2020-11-24	7227	1	0	2020-11-28	34	4
Total Rows: 2466 Query complete 00:00:00.262 CRLF Ln 2, Col 12											

Data Table

Data table as a source for exploratory data analysis

	A	B	C	D	E	F	G
1	country	customer_ID	invoice_amount_usd	disputed	dispute_lost	days_to_settle	
2	United Stat	3993-QUNVJ	7533	0	0	16	
3	Spain	8690-EEBEO	8333	0	0	41	
4	China	8156-PCYBM	7270	0	0	39	
5	China	6160-HCSFI	7272	0	0	22	
6	France	5284-DJOZO	5444	0	0	14	
7	Russia	5924-UOPGH	6760	0	0	28	
8	United Stat	3993-QUNVJ	5039	0	0	20	
9	Spain	6177-VTITE	6328	0	0	14	
10	France	8389-TCXFQ	7249	0	0	30	
11	China	3598-DNURW	5637	0	0	21	
12	China	6708-DPYTF	8674	0	0	46	
13	China	8156-PCYBM	6440	0	0	31	
14	United Stat	7228-LEPPM	3217	0	0	40	
15	France	2621-XCLEH	7951	0	0	46	
16	China	5196-TWQXF	6560	0	0	36	
17	France	6048-QPZCF	7392	0	0	5	
18	Spain	6391-GBFQJ	643	0	0	16	
19	Spain	7372-CESLR	2080	0	0	20	
20	United Stat	9323-NDIOV	4748	0	0	46	
21	United Stat	0625-TNJFG	4189	0	0	22	
22	China	2423-QOKIO	5642	0	0	21	
23	United Stat	3993-QUNVJ	6093	0	0	15	
24	China	6708-DPYTF	8031	0	0	42	
25	Spain	3271-YDPUJ	2822	0	0	27	
26	China	9841-XLGBV	4199	0	0	28	
27	United Stat	7228-LEPPM	2763	0	0	52	
28	Spain	1408-OQZUE	4508	0	0	36	
29	China	6160-HCSFI	7621	0	0	27	
30	United Stat	7260-ZHAKS	8298	0	0	36	
31	China	8156-PCYBM	4261	0	0	25	
32	United Stat	7228-LEPPM	4500	0	0	39	
33	Russia	5875-VZQCZ	8789	0	0	22	
34	France	9117-LYRCE	2209	0	0	41	
35	China	6708-DPYTF	9276	0	0	42	
36	China	8820-BLYDZ	7146	0	0	13	
37	United Stat	7260-ZHAKS	7283	0	0	27	
38	Russia	8364-UWVLM	6978	0	0	26	
39	Spain	7372-CESLR	3457	0	0	23	

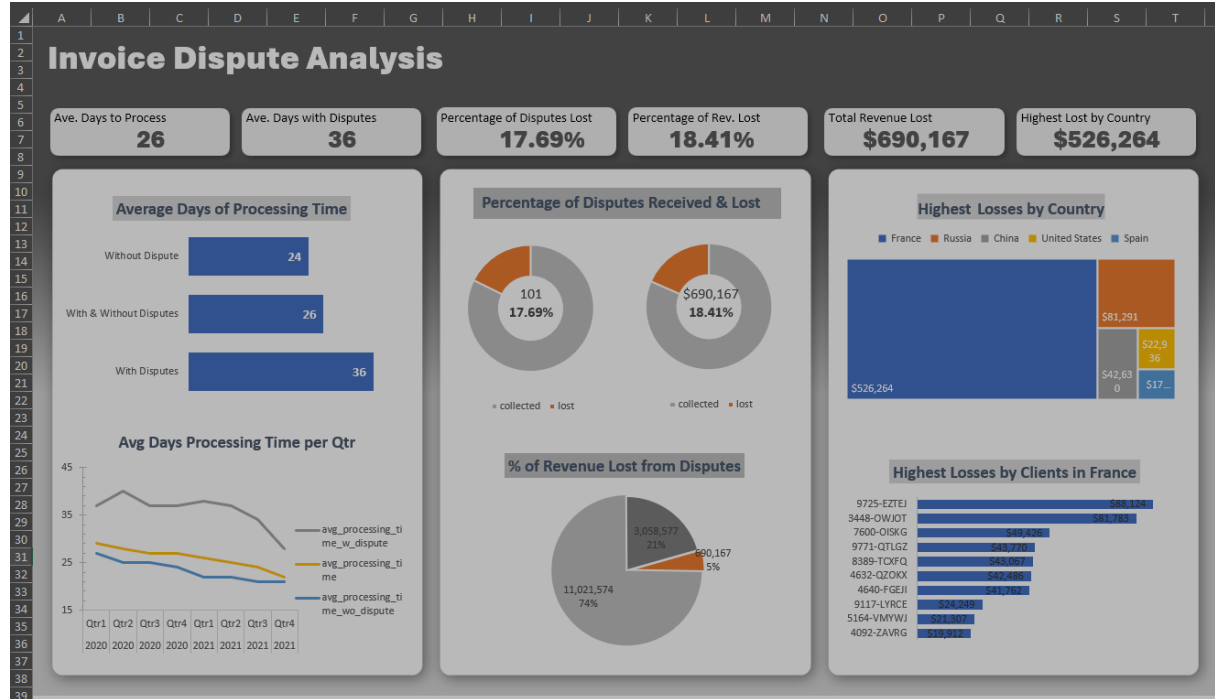
EDA

Exploratory data analysis using pivot tables



Excel Dashboard

A dashboard to summarize data report



SQL

A statistical SQL subquery and a common table expression (CTE)

The screenshot shows a SQL IDE interface. The top panel displays a SQL query using a Common Table Expression (CTE) and a statistical subquery. The query calculates the count and sum of invoice amounts, categorized by dispute status (won/lost), and then calculates the percentage of total counts and sums for each category.

```
1 WITH res AS (  
2     SELECT  
3         CASE  
4             WHEN dispute_lost = 0 THEN 'won'  
5             WHEN dispute_lost = 1 THEN 'lost'  
6             ELSE null  
7         END AS dispute_received,  
8         COUNT(dispute_lost) AS count_value,  
9         SUM(invoice_amount) AS amount_value  
10    FROM yell_inv  
11   WHERE disputed = 1  
12   GROUP BY dispute_received  
13 )  
14 SELECT  
15     dispute_received,  
16     count_value,  
17     amount_value,  
18     ROUND(count_value / SUM(count_value) over(), 4) AS count_percentage,  
19     ROUND(amount_value / SUM(amount_value) over(), 4) AS amount_percentage  
20 FROM res;
```

The bottom panel shows the 'Data Output' tab with a table of results. The table has five columns: dispute_received, count_value, amount_value, count_percentage, and amount_percentage. The data is grouped by dispute status (won/lost).

	dispute_received	count_value	amount_value	count_percentage	amount_percentage
1	won	470	3058577	0.8231	0.8159
2	lost	101	690167	0.1769	0.1841